

Antiderivatives and Indefinite Integrals

ANSWER KEY



Basic antiderivatives

- 1 Find each indefinite integral:

a $\int (3x^2 - 4x + 5) dx = x^3 - 2x^2 + 5x + C$

b $\int \frac{1}{x^2} dx = -\frac{1}{x} + C$

c $\int \sqrt{x} dx = \frac{2}{3}x^{3/2} + C$

d $\int \sec^2 x dx = \tan x + C$

- 2 Find $\int (4\sin x + e^x) dx$.
 $= -4\cos x + e^x + C$.

Using initial conditions

- 3 Find $f(x)$ given that $f'(x) = 6x^2 - 2$ and $f(1) = 5$.
 $f(x) = 2x^3 - 2x + C$. Using $f(1) = 5$: $2 - 2 + C = 5$, so $C = 5$: $f(x) = 2x^3 - 2x + 5$.
- 4 A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.
 $s(t) = \int v(t) dt = t^3 - 3t^2 + C$. Using $s(0) = 4$: $C = 4$: $s(t) = t^3 - 3t^2 + 4$.
- 5 An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s².
a Find the velocity function $v(t)$. $v(t) = -9.8t$ m/s (using $v(0) = 0$).
b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$. $s(t) = -4.9t^2 + 45$.
c Find how long it takes to hit the ground ($s = 0$), to two decimal places. $-4.9t^2 + 45 = 0$ gives $t = \sqrt{45/4.9} \approx 3.03$ s.

More antiderivatives

- 6 Find each indefinite integral:

a $\int (\frac{1}{x} - 3x^{-4}) dx = \ln|x| + x^{-3} + C$

b $\int (2\sec x \tan x) dx = 2\sec x + C$

c $\int \frac{4}{\sqrt{1-x^2}} dx = 4\sin^{-1} x + C$

- 7 Find the general antiderivative of $f(x) = (x + 1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).

Treating $(x + 1)$ as a single unit (power rule for a linear inner function): $\int (x + 1)^3 dx = \frac{(x + 1)^4}{4} + C$.

Motion applications

8 A rocket's acceleration is $a(t) = 6t \text{ m/s}^2$, with $v(0) = 0$ and $s(0) = 0$.

a Find $v(t)$. $v(t) = 3t^2$.

b Find $s(t)$. $s(t) = t^3$.

c Find the distance traveled in the first 4 seconds. $s(4) = 64 \text{ m}$.